

## VM172

### Stress Analysis of a Long, Thick, Isotropic Solenoid

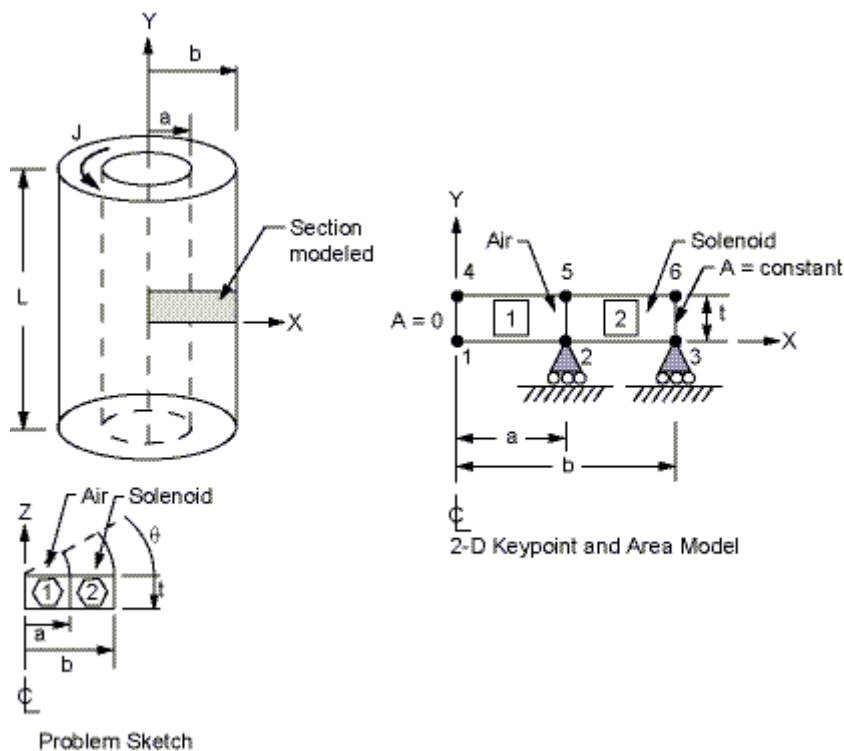
#### Overview

|                   |                                                                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference:        | F. C. Moon, <i>Magneto-Solid Mechanics</i> , John Wiley and Sons, Inc., New York, NY, 1984, pg. 275.                                                                                         |
| Analysis Type(s): | Coupled field Analysis ( <a href="#">ANTYPE</a> = 0)                                                                                                                                         |
| Element Type(s):  | 2-D Coupled-Field Solid Elements ( <a href="#">PLANE13</a> )<br>3-D Magneto-Structural Solid Elements ( <a href="#">SOLID62</a> )<br>3-D Magnetic Solid Elements ( <a href="#">SOLID97</a> ) |
| Input Listing:    | <a href="#">vm172.dat</a>                                                                                                                                                                    |

#### Test Case

A long, thick solenoid carries a uniform current density distribution,  $J$ . Assuming that the turns of the solenoid can be modeled as a homogeneous isotropic material with modulus of elasticity  $E$ , and Poisson's ratio  $\nu$ , determine the axial magnetic flux density distribution  $B_\theta$  and the circumferential stress  $\sigma_\theta$  distribution in the solenoid.

**Figure 172.1 Isotropic Solenoid Problem Sketch**



|                                                                                                         |                                                                                                                                    |                                                     |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Material Properties</b><br>$E = 10.76 \times 10^{10} \text{ N/m}^2$<br>$\nu = 0.35$<br>$\mu = \mu_0$ | <b>Geometric Properties</b><br>$a = .01 \text{ m}$<br>$b = .02 \text{ m}$<br>$t = .002 \text{ m}$<br>$\Theta = 10 \text{ degrees}$ | <b>Loading</b><br>$J = 1 \times 10^6 \text{ A/m}^2$ |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|

#### Analysis Assumptions and Modeling Notes

The problem is solved first using a 2-D axisymmetric analysis with [PLANE13](#) elements and then a 3-D analysis using [SOLID62](#) and [SOLID97](#) elements. The length of the solenoid is assumed infinite ( $L = \infty$ ), so only a section of the axisymmetric solenoid ( $t = .002$  m, arbitrary) is required for modeling. It is assumed that the magnetic field external to the solenoid is zero, so the nodes at  $x = b$  are coupled ( $AZ = \text{constant}$  (2-D),  $AY = \text{constant}$  (3-D)) such that the proper flux-parallel boundary condition is imposed. The flux-parallel condition at  $x = 0$  is imposed by setting  $A = 0$ . Flux-normal boundary conditions are imposed naturally (no ANSYS input necessary) at  $y = 0$  and  $y = t$  for the 2-D analysis, while in 3-D, the normal component ( $AZ$ ) is set to zero. In 3-D, the nodes are rotated into cylindrical coordinates so that all in-plane vector potentials ( $AX$  and  $AZ$ ) can be set to zero since only  $AY$  is required for a true axisymmetric field solution.

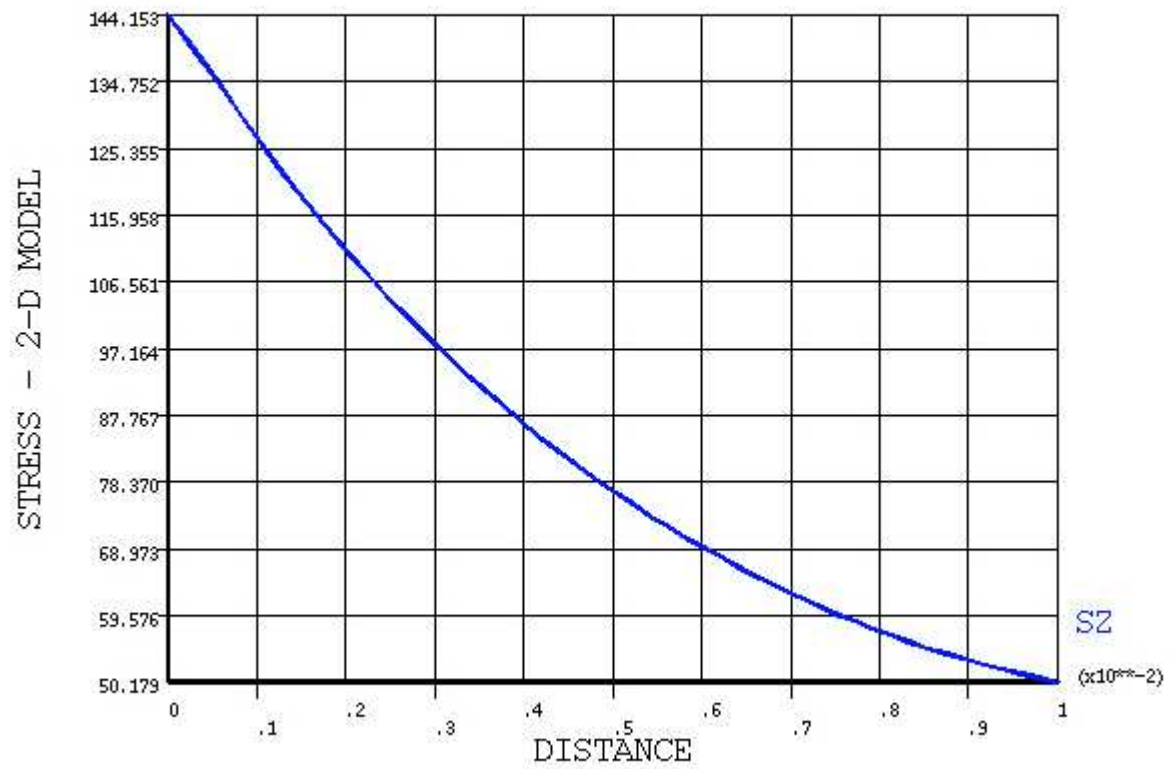
Symmetric structural boundary conditions are applied to the solenoid elements at  $y = 0$ . The nodes at  $y = t$  on the solenoid are coupled in  $UY$  to ensure symmetry. The air is modeled with 5 elements in the radial direction while the solenoid is discretized with 20 elements in the radial direction to accurately model the stress distribution. The 3-D model uses [SOLID97](#) elements to model the air and [SOLID62](#) elements to model the coil.

#### Results Comparison

|                                                   |                                                       | Target <sup>[1]</sup> | ANSYS   | Ratio |
|---------------------------------------------------|-------------------------------------------------------|-----------------------|---------|-------|
| <a href="#">PLANE13</a>                           | B <sub>angle</sub> , T @ $r = .01$ m                  | .01257                | .01226  | 0.975 |
|                                                   | B <sub>angle</sub> , T @ $r = .013$ m                 | .008796               | .008797 | 1.000 |
|                                                   | B <sub>angle</sub> , T @ $r = .017$ m                 | .003770               | .003769 | 1.000 |
|                                                   | Stress <sub>o</sub> , N/m <sup>2</sup> @ $r = .01$ m  | 146.7                 | 144.2   | 0.983 |
|                                                   | Stress <sub>o</sub> , N/m <sup>2</sup> @ $r = .013$ m | 97.79                 | 97.70   | 0.999 |
|                                                   | Stress <sub>o</sub> , N/m <sup>2</sup> @ $r = .017$ m | 62.44                 | 62.61   | 1.003 |
| <a href="#">SOLID62</a> / <a href="#">SOLID97</a> | B <sub>angle</sub> , T @ $r = .01$ m                  | .01257                | .01219  | 0.970 |
|                                                   | B <sub>angle</sub> , T @ $r = .013$ m                 | .008796               | .008747 | 0.994 |
|                                                   | B <sub>angle</sub> , T @ $r = .017$ m                 | .003770               | .003750 | 0.995 |
|                                                   | Stress <sub>o</sub> , N/m <sup>2</sup> @ $r = .01$ m  | 146.7                 | 144.4   | 0.984 |
|                                                   | Stress <sub>o</sub> , N/m <sup>2</sup> @ $r = .013$ m | 97.79                 | 95.97   | 0.981 |
|                                                   | Stress <sub>o</sub> , N/m <sup>2</sup> @ $r = .017$ m | 62.44                 | 61.34   | 0.982 |

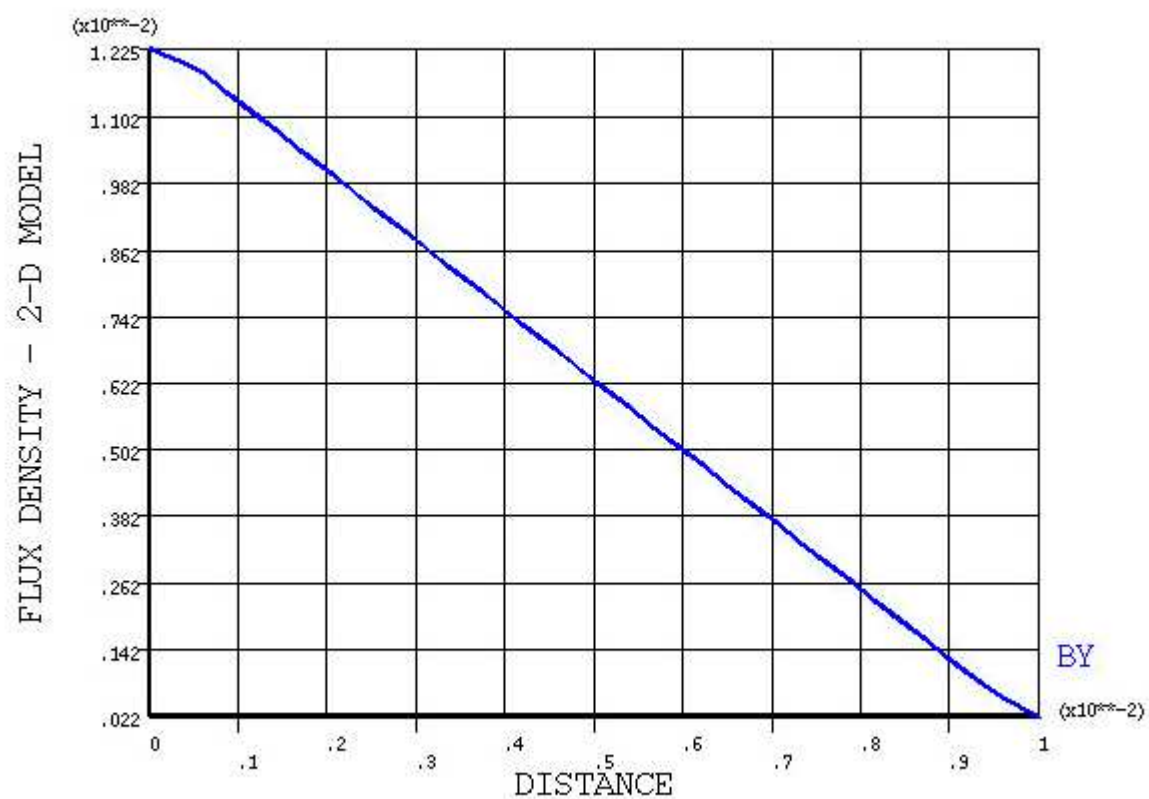
1. Assumed to be linearly varying through solenoid

**Figure 172.2 2-D Circumferential Stress through Solenoid Windings**



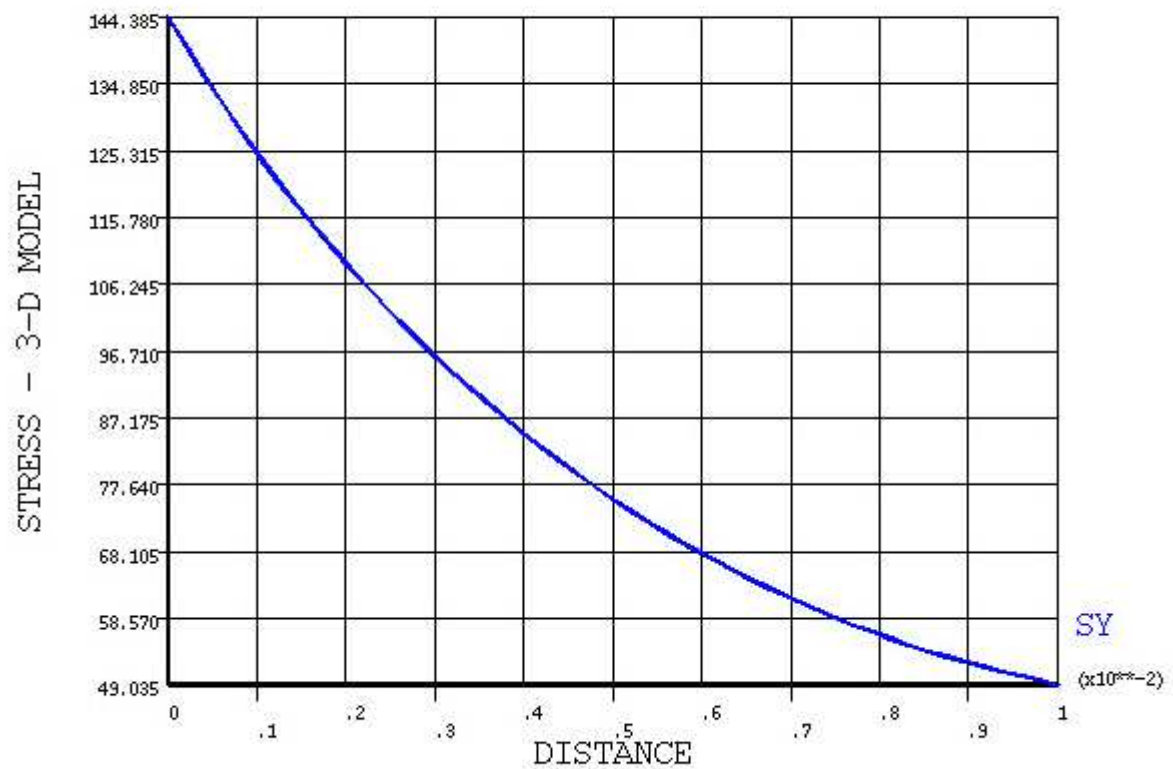
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Figure 172.3 2-D Axial Flux Density through Solenoid Windings



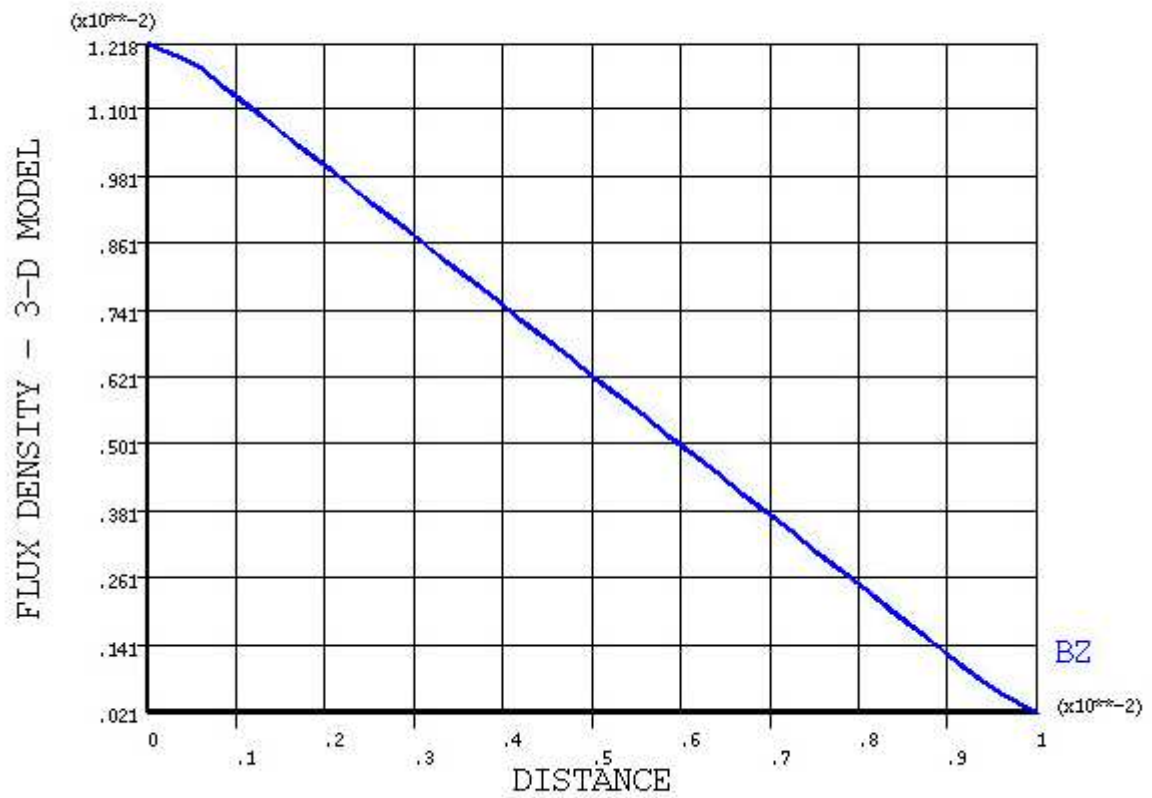
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Figure 172.4 3-D Circumferential Stress through Solenoid Windings



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Figure 172.5 3-D Axial Flux Density through Solenoid Windings



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